

Stablecoin Stability Analysis

Predicting Depeg Risk with Machine Learning

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Why Stablecoin Stability Matters

The Problem:

- Stablecoins are designed to maintain a \$1.00 peg
- **Depeg events** cause massive financial losses
- UST/LUNA crash (May 2022): \$40B+ wiped out
- USDC depeg (March 2023): dropped to \$0.87

Our Goal:

- Detect anomalous price behavior early
- Predict stability risk 1 hour ahead
- Understand market regime patterns

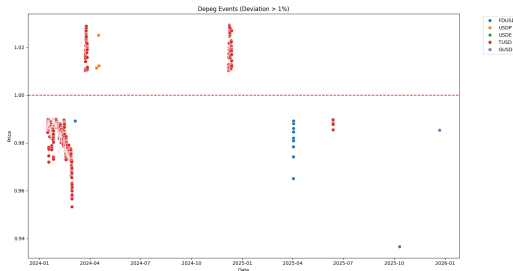


Figure: Historical depeg events (deviation >1%).

Data Collection & Scope

Data Sources:

- **Coinbase:** USDT/USD, DAI/USD
- **Binance:** TUSD, FDUSD, USDP, USDE (vs USDT)
- **Context:** BTC/USD, ETH/USD for correlation

Dataset Statistics:

- **Time Range:** ~2 years hourly data
- **Records:** 89,003 observations
- **Train/Test:** 70,792 / 18,211
- **Depegs:** 1,271 events ($>1\%$)

Assets Processed:

Coin	Exchange	Status
USDT, DAI	Coinbase	✓
TUSD, FDUSD	Binance	✓
USDP, USDE	Binance	✓
BUSD/FRAX/GUSD	–	Skipped

Skipped assets had <600 rows of data.

Feature Engineering: 34 Predictive Features

Price Features:

- Deviation from \$1 (abs & %)
- 24h rolling volatility
- Price momentum (1h/6h/24h)
- Deviation velocity (rate of change)

Volume Features:

- 24h moving average
- 24h standard deviation
- Volume velocity
- Volume z-score

Lag & Context Features:

- Multi-hour lags (1, 2, 3, 6, 12, 24h)
- BTC/ETH prices & returns
- 24h rolling correlations with BTC/ETH
- Deviation z-score

Prediction Targets

- **Classification:** Binary risk label – “Low” (depeg risk $>1\%$) vs “Not-Low”
- **Regression:** Continuous deviation percentage 1 hour ahead

Methodology: Four ML Approaches

1. Unsupervised Learning:

- **Clustering:** Identify market regimes
 - K-Means (3 clusters)
 - DBSCAN (density-based)
- **Anomaly Detection:** Flag potential depegs
 - Isolation Forest
 - Local Outlier Factor (LOF)

2. Supervised Learning:

- **Classification:** Predict binary risk
 - Random Forest (500 trees)
 - Gradient Boosting (300 estimators)
- **Regression:** Predict deviation magnitude
 - Ridge Regression (weighted)
 - Neural Network (MLP: 128-64)

Handling Class Imbalance

Only 1.5% of samples are “Low” (depeg) class. We applied: (1) 10× oversampling of Low class, (2) class weights (1:20), (3) aggressive probability threshold (0.005).

Exploratory Analysis: Price Stability Varies by Stablecoin

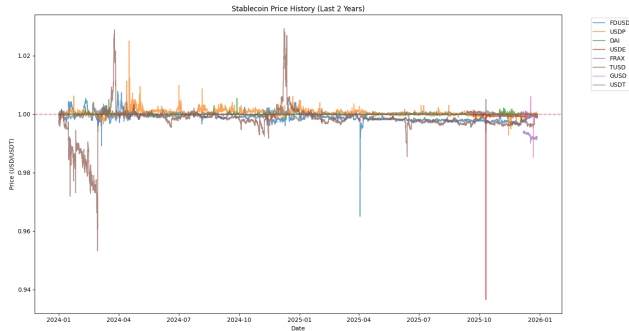


Figure: 2-year price history across all stablecoins.

Key Observations:

- **USDT & DAI:** Extremely stable, rarely deviate $>0.1\%$
- **TUSD:** Most volatile – major depeg events visible
- **FDUSD/USDP/USDE:** Moderate fluctuations

Implication:

- Models must handle both stable and volatile regimes
- Per-symbol evaluation critical for fair assessment

Exploratory Analysis: Cross-Asset Correlations

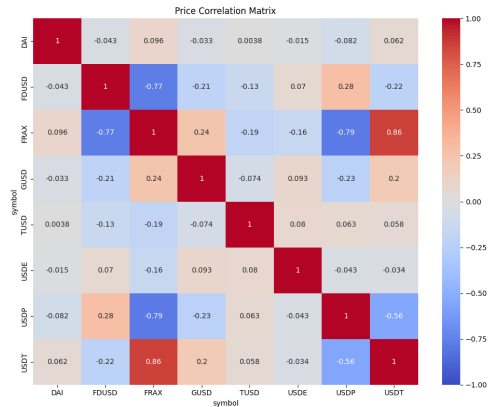


Figure: Price correlation matrix.

Correlation Insights:

- Most stablecoins show **low correlation** with each other (independent behavior)
- **BTC & ETH** are highly correlated (0.9+) as expected
- Stablecoin-crypto correlations are weak
 - Suggests depegs are *not* purely driven by market crashes
 - Idiosyncratic factors (issuer risk, liquidity) matter

Feature Engineering Value:

Including BTC/ETH context helps capture market-wide stress events that may precede depegs.

Clustering: Identifying Market Regimes

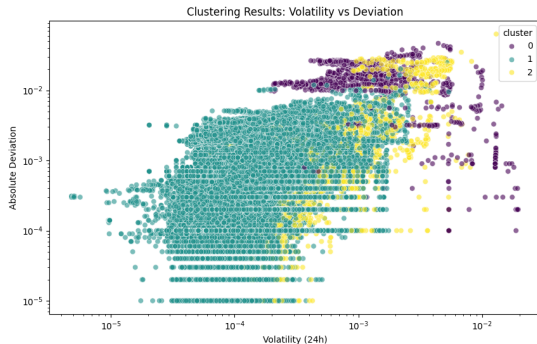


Figure: K-Means clusters: Volatility vs Deviation.

Model Comparison:

Metric	K-Means	DBSCAN
Clusters	3	378
Silhouette	0.490	0.234
Noise Points	0	84,933

3 Regimes Identified:

- **Stable:** Low volatility, minimal deviation
- **Moderate:** Elevated volatility
- **Stressed:** High volatility, large deviations

Winner: K-Means – 2× better silhouette.

Classification: Predicting Depeg Risk (1h Ahead)

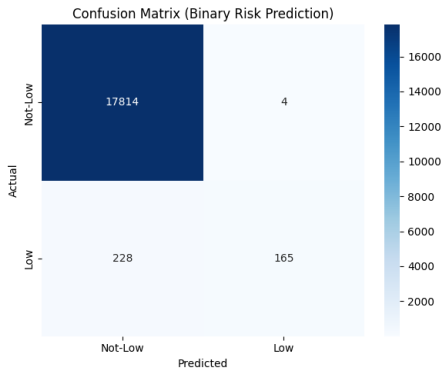


Figure: Confusion matrix (Random Forest).

Model Comparison (Test Set: 18,211 samples):

Metric	Random Forest	Gradient Boosting
Accuracy	98.73%	98.61%
F1 (weighted)	0.985	0.983
Low Precision	98%	99%
Low Recall	42%	36%
Low F1	0.59	0.53

Per-Symbol Accuracy:

DAI: 100% — USDT: 100% — FDUSD: 99.9% — USDP: 99.9%
— USDE: 99.8% — TUSD: 94.2%

Winner: Random Forest – 17% better recall on critical “Low” class (165 vs 142 true positives captured).

Anomaly Detection: Flagging Depeg Events

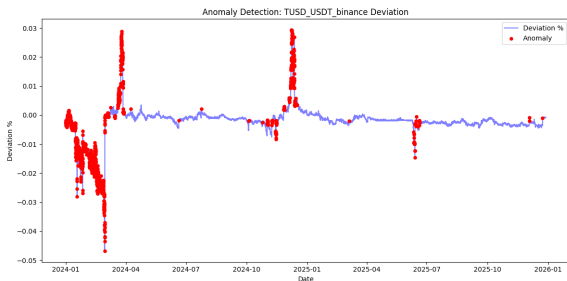


Figure: Isolation Forest anomalies on TUSD (most volatile).

Ground Truth: 1,271 depeg events ($>1\%$ deviation)

Model Comparison:

Metric	Isolation Forest	LOF
Detected	2,227	2,977
Precision	0.474	0.118
Recall	0.831	0.277
F1 Score	0.604	0.166

Winner: Isolation Forest

- Captures **83%** of true depegs
- $3.6\times$ better F1 than LOF
- Per-symbol contamination tuning improved results

Regression: Forecasting Deviation Magnitude (+1h)

Global Performance:

Metric	Ridge	Neural Net
RMSE	0.00081	0.00149
R ²	0.951	0.834

Per-Symbol R²:

Symbol	R ²	Symbol	R ²
TUSD	0.987	FDUSD	0.738
USDT	0.911	USDP	0.557
DAI	-0.024	USDE	-0.725

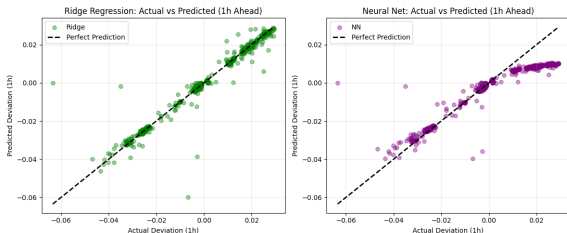


Figure: Predicted vs actual deviation (Ridge & NN).

Winner: Ridge Regression – 1.8× lower RMSE, simpler model generalizes better.

Key Findings & Conclusions

Model Performance:

- **Clustering:** K-Means finds 3 market regimes
- **Classification:** RF achieves 98.7% accuracy, 42% recall on depegs
- **Anomaly:** IsoForest captures 83% of depegs ($F1=0.60$)
- **Regression:** Ridge achieves $R^2=0.95$

Insights:

- TUSD shows frequent depegs; DAI/USDT are stable
- Volatile coins easier to forecast (TUSD: $R^2=0.99$)
- Class imbalance requires aggressive handling
- Simpler models win: Ridge > NN, K-Means > DBSCAN

Practical Application

IsoForest + Ridge can form a real-time alert system: flag anomalies (83% recall), then quantify expected deviation ($R^2=0.95$).

Technical Summary

Implementation:

- **Data:** ccxt for exchange APIs
- **ML:** scikit-learn (K-Means, RF, Ridge, IsoForest, MLP)
- **Viz:** matplotlib, seaborn, plotly

Codebase:

- `src/data_collector.py`
- `src/preprocessor.py`
- `src/modeling.py`
- `src/visualization.py`

Final Metrics:

Task	Best Result
Clustering	Silhouette 0.490
Classification	Acc 98.7%, F1 0.59
Anomaly	F1 0.60, Recall 83%
Regression	R^2 0.95, RMSE 0.0008

Best Models: K-Means, Random Forest, Isolation Forest, Ridge Regression

Thank You!

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